UNFORGED · MIDNIGHT BUILDATHON · WAVE 2

Evidence without exposure.



Unforged proves that digital evidence came through an authorized capture and editorial workflow, without exposing the person or device behind it. A source-safe privacy and authorization layer for C2PA, built on Midnight.

`github.com/bunnyyxtan/unforged` | `Compact 0.35 · ZKIR v3` | `34 tests · CI green`

Provenance needs a trusted signer. Naming the signer can get them jailed.

330

journalists imprisoned at the end of 2025 (CPJ prison census)

129

journalists and media workers killed in 2025, CPJ's highest annual total

The default today

Public signatures build durable identity and activity graphs. Centralized verifiers see every signer-to-file link. C2PA itself lists privacy, harms and human rights as design goals.

Newsrooms must choose: authentic and traceable, or anonymous and unverifiable.

Unforged removes the choice: the public record proves the file came from an authorized pool and passed a defined workflow, and withholds who approved it.

Sources: CPJ 2025 prison census; CPJ 2026 report on 2025 killings; C2PA 2.3 specification. Links in docs/RESEARCH.md.

WHY THIS NEEDS MIDNIGHT

Verification stays public. Authorization stays private.

	CENTRAL SERVICE	PUBLIC CHAIN	C2PA ALONE	UNFORGED
Proves organizational authority	trust vendor	yes	yes	in ZK
Hides which member signed	operator sees	key on-chain	signer cert	yes
Hides which admins approved	no	no	no	nullifiers
Independently checkable	no	yes	trust list	yes
No tokens for contributors	yes	wallet	yes	sponsored

Public on the ledger

File fingerprint • hour-window capture time • pool epoch • opaque commitments • approval tally and nullifiers

Never leaves the prover

P-256 key and salt • passkey signature and authenticator data • Merkle path • approver secret • contributor identity

Without private execution, a provenance system either publishes credentials or trusts a central verifier. Compact verifies the P-256 signature and two Merkle memberships inside one proof and discloses only what the contract selects.

THE PLATFORM · WAVE 2 BUILD

Six products. One trust boundary.

LIVE · L0-L1

Capture

Passkey or labeled test credential approves the exact bytes. Native attestation and C2PA capture next.

LIVE

Desk

Every seal becomes a case: evidence matrix, checklist-gated approval, hold reasons, decision log, publication package.

LIVE · 2 OF 3

Trust Pools

Anonymous threshold enrollment in circuit, revocation epochs, anonymity floor. The owner key alone cannot enroll.

LIVE

Verify

Eight-row security report: each claim labeled established, asserted, reviewed, unresolved or unavailable.

LIVE · LOCAL

Relay

Batches of three, shuffled submission order, capture time published to the hour.

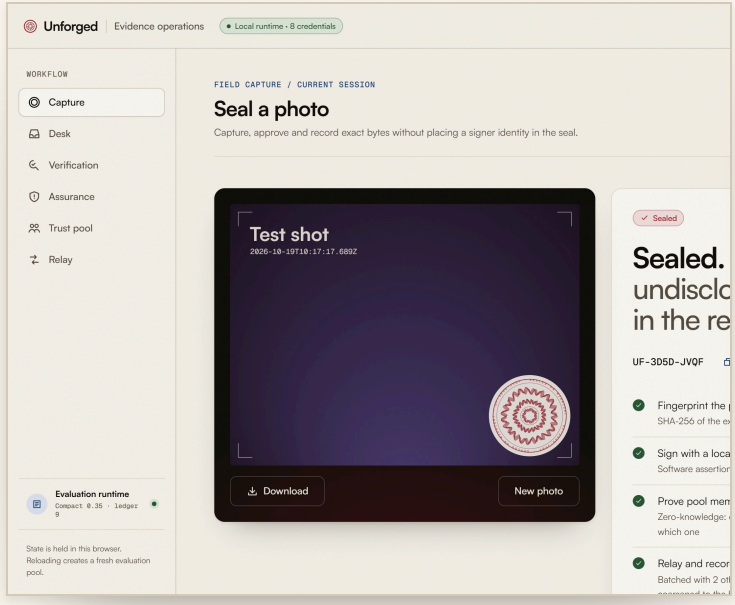
DRAFT PROFILE

Platform

Draft C2PA `org.unforged.authorization` assertion and an offline CLI verifier. API, SDK and CMS plugins follow.

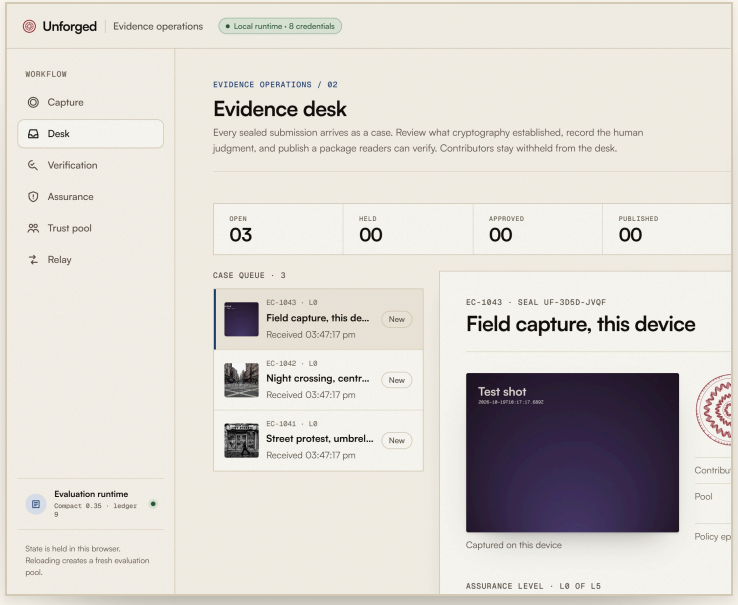
ONE FILE THROUGH THE WORKFLOW

Capture. Review. Publish what readers can check.



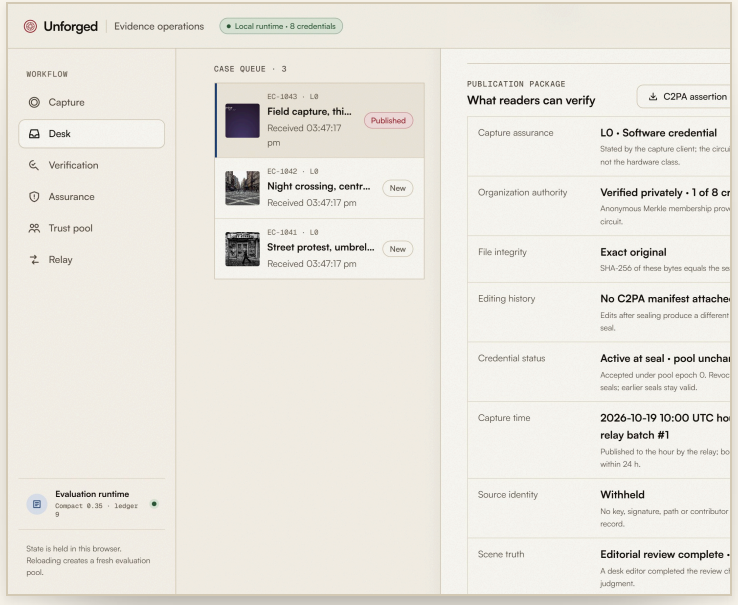
01 • Capture

2-of-3 approved credential seals the exact bytes; relay batch #1, hour window.



02 • Desk

A case with assurance level, withheld contributor and a checklist before approval.



03 • Publish and verify

Every claim says how it is known. Scene truth is a human review, never a crypto claim.

A passkey signature and two anonymous memberships, verified inside one proof.

CIRCUIT	PRIVATE INPUT	GUARANTEE
`seal`	P-256 key, salt, Merkle path, WebAuthn assertion	Anonymous membership + in-circuit `secp256r1EcdsaVerify` over pool and fingerprint; UP+UV; uniqueness; 24 h window; pool floor
`approve`	Approver secret, approver path	Anonymous approver membership; nullifier per (approver, credential); k-th approval enrolls
`enroll`	Owner secret	Direct enrollment only when no threshold policy exists
`addApprover`	Owner secret	Opaque approver commitment
`revoke`	Owner secret	Removes a credential, resets accepted roots, bumps the epoch

34

simulator tests with real P-256 WebAuthn assertions

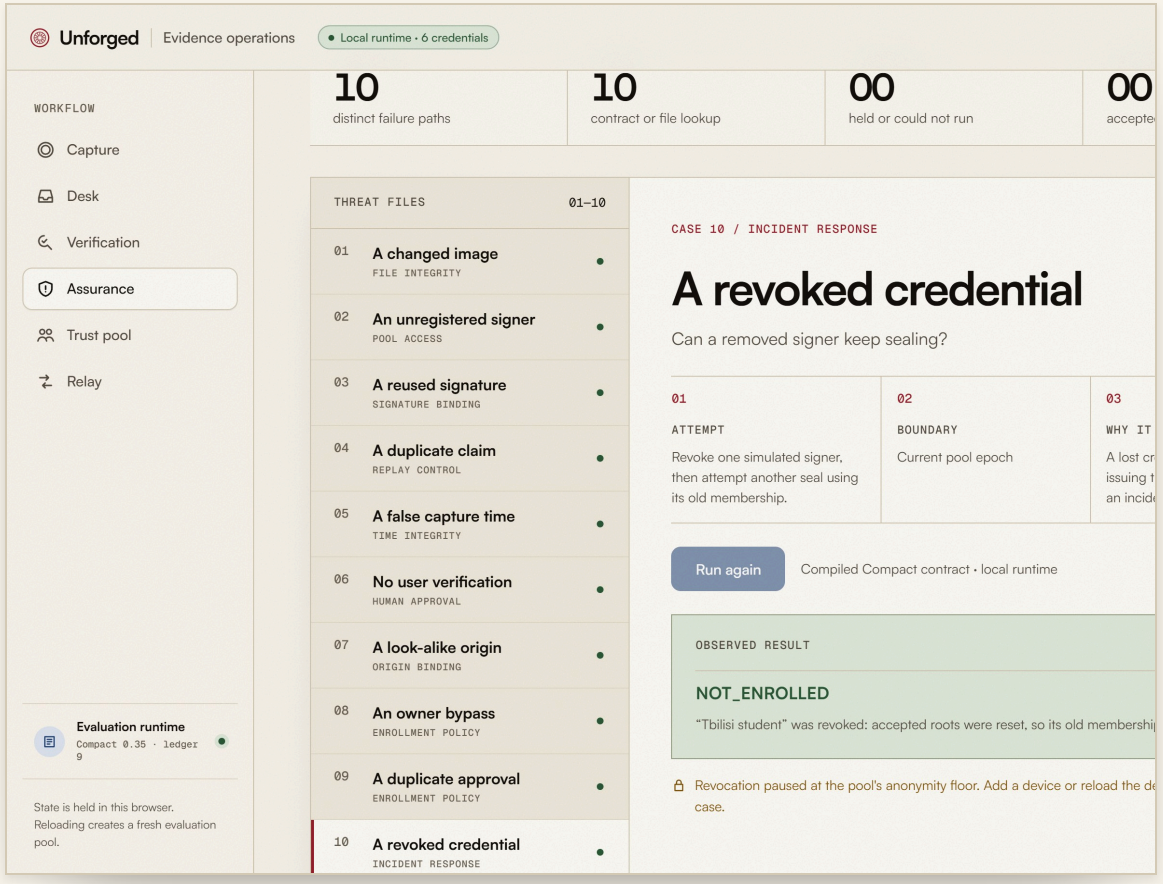
0

bytes of public state differ, whichever member sealed or approvers approved

Compact 0.35 • `--feature-zkir-v3` • ledger 9 • runtime 0.20. WebAuthn profile adapted from Midnight Passport (Apache-2.0). CI compiles the contract, runs every test, typechecks and builds the app on every commit. The React workspace executes the same generated circuits.

Refusals are product behavior.

ATTEMPT	OBSERVED
Change one pixel of a sealed file	NO_SEAL
Credential outside the pool	NOT_ENROLLED
Reuse a signature for other pixels	BAD_SIGNATURE
Second signer claims the same file	ALREADY_SEALED
Backdate capture by two days	BAD_TIME
No user verification	UV_REQUIRED
Look-alike origin	BAD_SIGNATURE
Owner enrolls without approvers	APPROVAL_REQUIRED
One approver approves twice	ALREADY_APPROVED
Seal after revocation	NOT_ENROLLED



HONEST ASSURANCE

A browser passkey is not a Sony camera. The verifier says so.

L0	L1	L2	L3	L4	L5
Software test credential	WebAuthn approval, UP + UV	Device-bound hardware key	Hardware and app attestation	C2PA capture credential	Sensor evidence and trusted time

Established in Wave 2

- Exact-file integrity and private pool authority
- Credential active at the sealing epoch; anonymous k-of-n enrollment
- Signer and approvers withheld from the record
- Capture time bounded by ledger time, published to the hour

Not claimed yet

- Camera-sensor origin (browser capture can be modified)
- Public-network inclusion: ledger 9 is not on Preprod yet, so the contract runs on the local ledger-9 runtime
- Network-level anonymity: the relay runs in the browser today
- Signed C2PA manifests: the assertion is a draft profile

COMPANY

Start where exposure is lethal. Scale through platforms.

Beachhead

Newsrooms, fact-checkers and human-rights teams that receive media from freelancers, stringers and sensitive sources. Buyer: security, standards or photo-desk leadership.

Expansion

Election monitoring → insurance and legal evidence → science and supply-chain field records → marketplaces → AI training-data provenance → cameras, phones and platforms.

Revenue layers

Enterprise platform (hypothesis: $1—3k/month small newsroom, $25—150k/year national) • Verify API • Capture SDK • self-hosted • OEM licensing • assurance services.

Moat

Private organizational authorization wired into C2PA, hardware and CMS workflows, plus an anonymity network effect: larger compatible pools protect every contributor better.

First-year targets

3—5 design partners • 50—100 protected contributors • 10,000 real captures • 2 CMS integrations • first paid annual contract. Prices and targets are hypotheses until pilots.

WAVE 3 · 27 OCT – 16 NOV

From a working platform to network evidence.

Network

Real ledger-9 proofs, public deployment when supported, independent indexer checks, sponsored fees.

Capture

Native Android prototype with hardware-backed keys and attestation (L2—L3).

Interop

Signed C2PA manifests carrying the authorization assertion; public seal pages.

Operations

Persistent multi-user Desk with roles, independent relays, 1—3 design partners.

The evidence can be public.
The witness does not have to be.

`github.com/bunnyyxtan/unforged` `Built on Midnight` `Apache-2.0`